School of Informatics & Cybersecurity - Assessment Submission Coversheet

Student Information

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Module Information

Module Title:	Fundamentals of AI & Data Science
Module Code:	COMP H2706

Assessment Information

Assessment Title:	Project

Declaration

☒ I confirm that I have read and understood the University's Academic Integrity Policy.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/academic-quality-assurance-and-enhancement/academic-integrity/

☒ I have completed the TU Dublin AI Driving License theory test.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/lta/lta-resources/artificial-intelligence-driving-license/

☒ I declare that this assessment submission is my own work.

☒ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

Student signature:	Róisín Aylward
Date:	02/04/2026

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Executive Summary

This report presents an analysis of a cybersecurity network traffic dataset for a financial organisation that has experienced costly cyberattacks. Following the CRISP-DM framework, two machine learning tasks were completed: classifying network sessions as Benign or BruteForce (attack_category), and predicting average packet size (avg_packet_size). The dataset originally contained 6,000 session records across 19 features covering login behaviour, traffic volume, connection patterns, and network protocol information. After data preparation, there were 5,856 rows remaining. Two classification models were trained, a Decision Tree (baseline) and KNN. Two prediction models were also trained: Linear Regression (baseline) and a Decision Tree Regressor. The Decision Tree achieved higher recall of 0.686, making it the more suitable classifier for the business where missing a real attack is far more costly than a false alarm, though KNN had a marginally better F1-score of 0.5923. For the prediction models, Linear Regression outperformed the Decision Tree Regressor with an R2 of 0.548. The top features for attack detection were packet_rate, bytes_ratio (added), signal_strength, anomaly_score, and bytes_received.

Introduction

The financial sector is a target for cybercrime. Network intrusions usually leave patterns in traffic before any real damage can be done. A common threat is the brute force attack, which IMB defines as "a type of cyberattack in which hackers try to gain unauthorized access to an account or encrypted data through trial and error, attempting several login credentials or encryption keys until they find the correct password." Using Python and data science principles such as the CRISP-DM framework, we can analyse network session logs to identify patterns associated with brute force attacks, to help detect and prevent future attacks from occurring.

The dataset used captured individual network sessions as rows. The columns represent attributes represents network identifiers such as the whether the protocol was TCP, UDP or ICMP, the TCP flag, and the destination port number of the connection. Other attributes that are represented are the total amount of login attempts and failed login attempts in a session as well as traffic volume measurements such as the amount of bytes sent, bytes received, packet rate that measures the number of packets transmitted per second, and average packet size which is the Prediction Target. The attack category which is the Classification Target either labels the session as a brute force attack or as benign, and the anomaly score is a risk score ranging from 0 to 1 to indicate how suspicious a session is.

Data Understanding & Exploration

The Dataset contained 6,000 rows/sessions and 19 columns/attributes. No duplicate rows were found. Three columns had missing values, each exactly 420 which amounts to 7%, two of which are numeric: bytes received and signal strength, and the third being protocol which is categorical. Another two columns were found to have contained some values that are impossible. Firstly, there were 1,109 rows (or 18.483%) where failed logins were mis-recorded to exceed total login attempts. Secondly, 144 rows (or 2.4%) of the session retries were also mis-recorded to exceed the total connection count.

The classification target attack category's two classes are split with 4,161 sessions (or 69.4%) being labelled as "Benign", and 1,839 sessions (30.6%) labelled as "BruteForce", making the dataset imbalanced. As a result, a model that always predicts Benign would get a 69.4% accuracy while detecting zero attacks. In Cybersecurity, a false negative by predicting a BruteForce session as Benign and missing an attack is more harmful than a false positive of predicting a Benign session as BruteForce and causing a false alarm. Therefore, a high recall, which measures the model's ability to detect all actual BruteForce cases, and would mean fewer missed attacks, was used as the primary evaluation metric. To reduce the risk of alarm fatigue, F1-Scores, which balances the model's precision scores on correctly predicting BruteForce sessions with the model's recall scores, were also considered when evaluating models. It is important to note that the proportions of network sessions being Benign and BruteForce does not correspond to the proportion of legitimate users and malicious actors as MITRE ATT&CK states that a single BruteForce attack may use more than one network session.

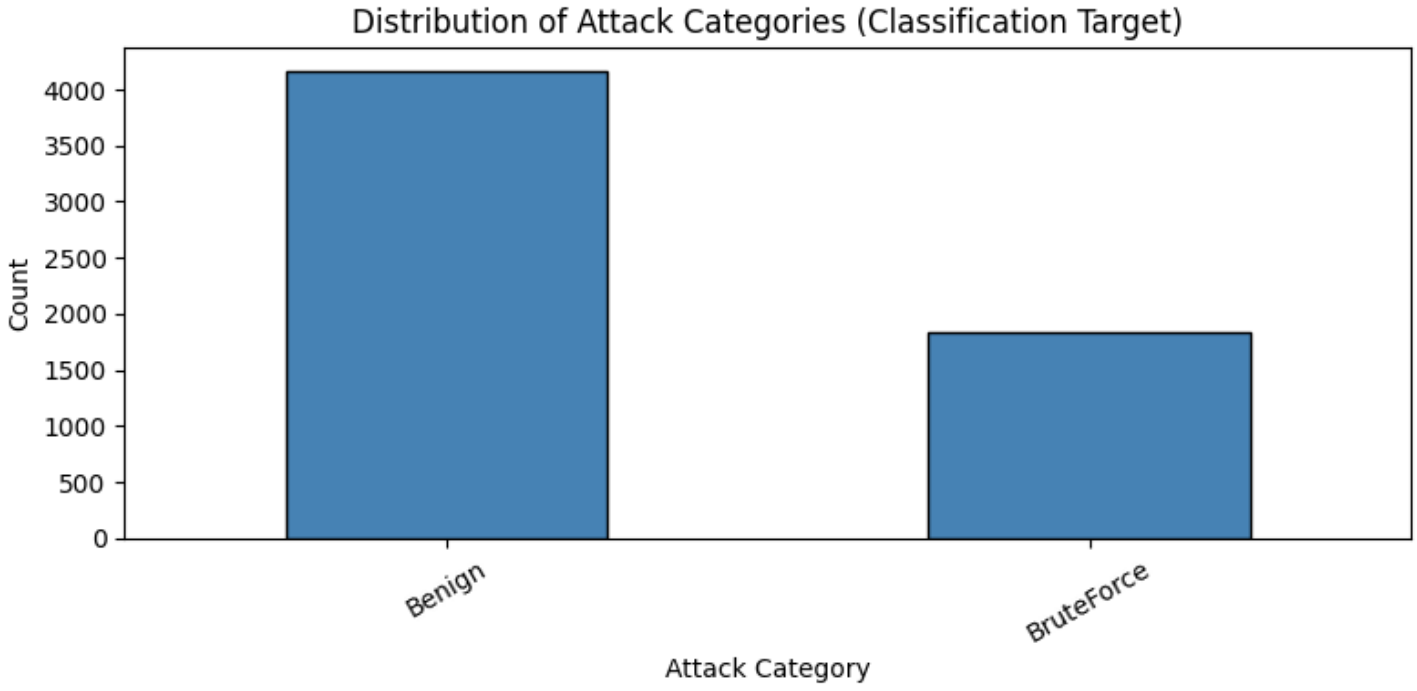


Figure 1 above: Classification Target Distribution bar chart

The prediction target average packet size has a symmetric distribution as evidenced by the histograms in figure 3 and the almost identical mean of 291.96 bytes and median of 291.74 bytes. The range spans from 198.96 bytes to

375.36 bytes. Boxplots by attack category showed nearly identical distributions for Benign and BruteForce sessions, suggesting average packet size alone has very little predictive power for distinguishing attack types.*

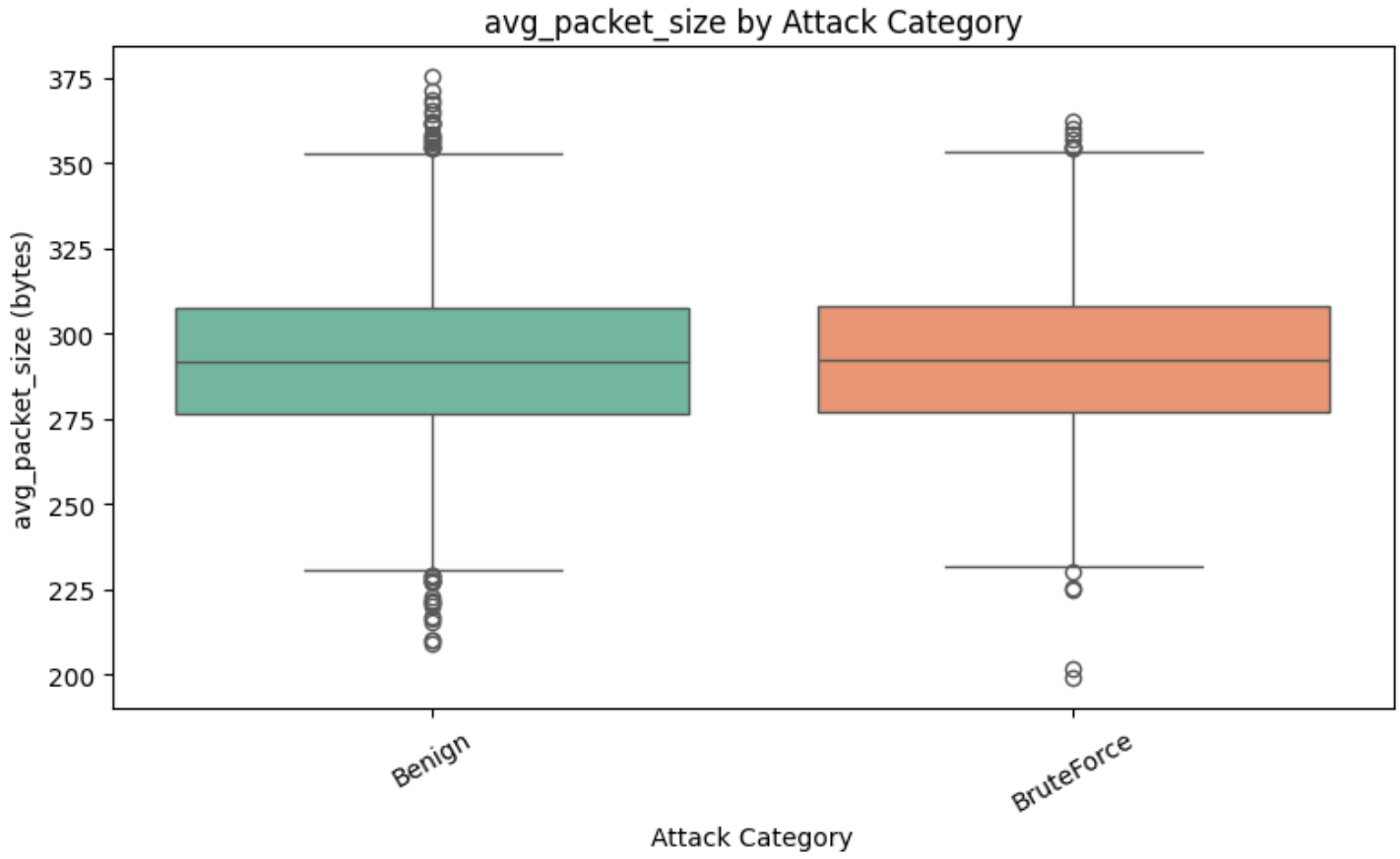


Figure 2 above: avg_packet_size by attack category boxplot

Histograms of all numeric columns showed most features are roughly symmetric. Bytes sent was the most right-skewed, consistent with most sessions transferring modest amounts but a small number transferring very large payloads. This is a pattern often associated with data exfiltration in real networks (source).

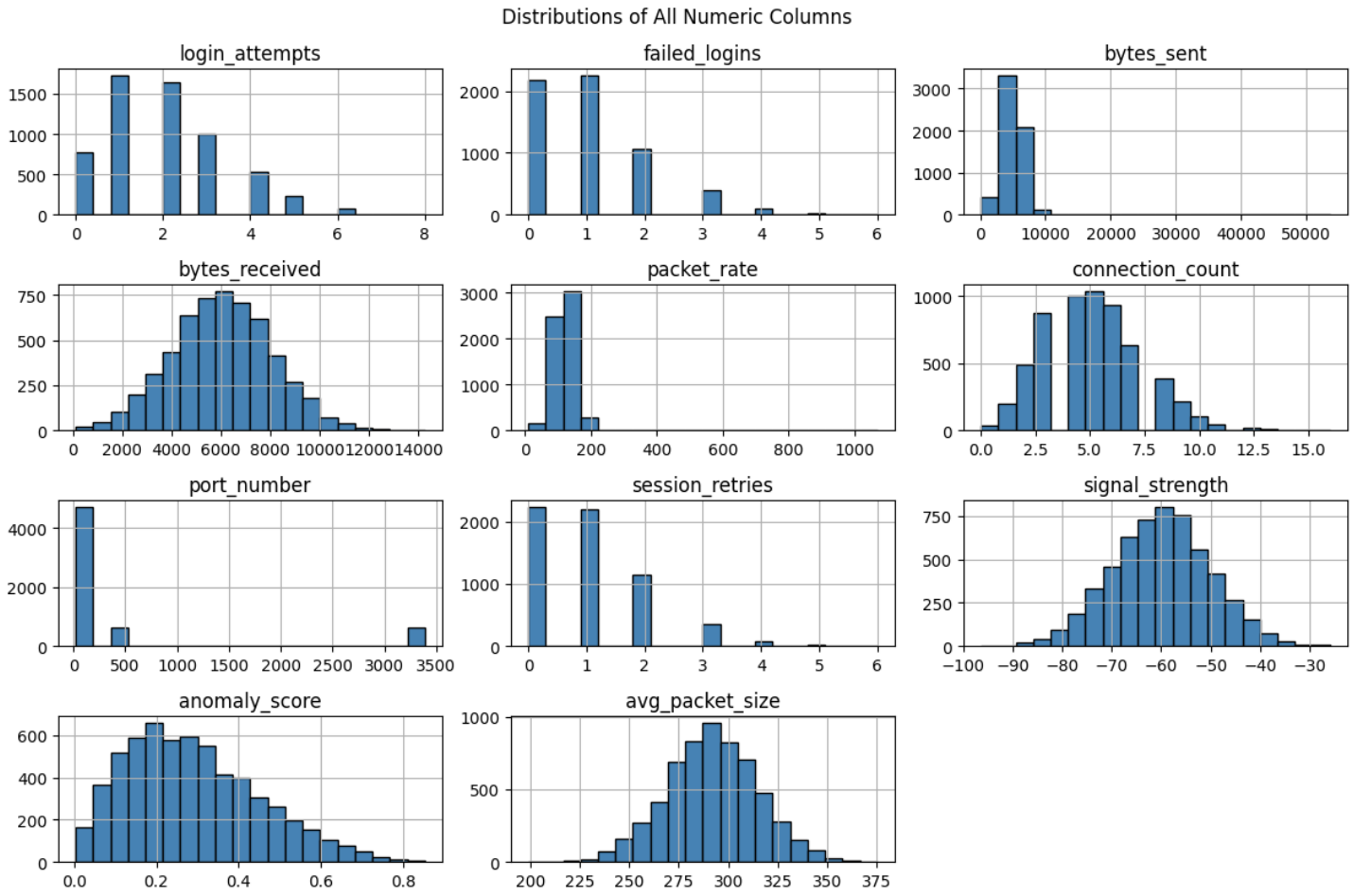


Figure 3 above: Histograms of all numeric columns

Boxplots confirmed outliers across several columns. The most notable were port number, which had 1,265 values outside the IQR bounds, and session retries with 375 outliers. These were handled with IQR capping rather than row removal in the data preparation stage.

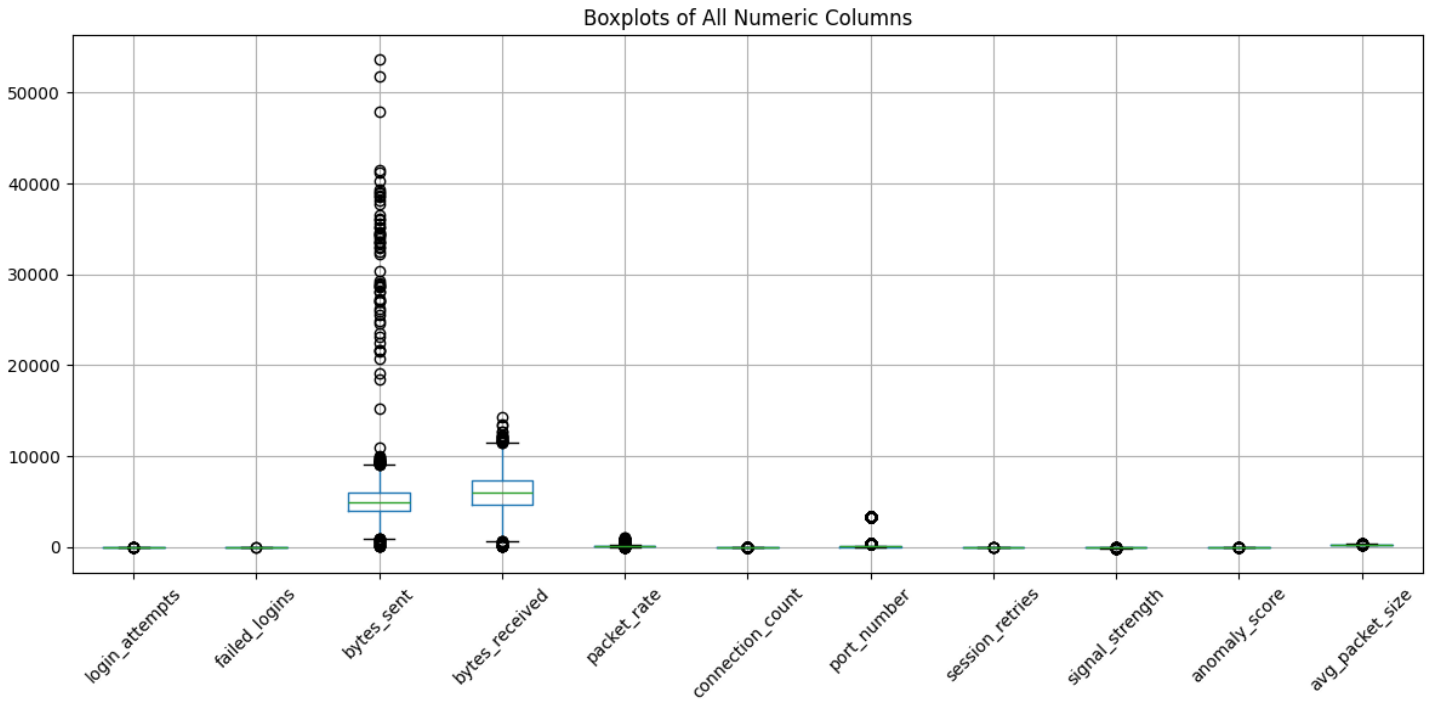


Figure 4 above: Boxplots of all numeric columns

Checking value counts before standardising showed a data quality issue in the protocol column. Both uppercase (TCP, UDP) and lowercase (tcp, udp) versions of the same values existed. After lowercasing all text columns during data preparation, protocol merged into three categories: 3,002 rows with tcp, 1,983 with udp, and 595 with icmp.

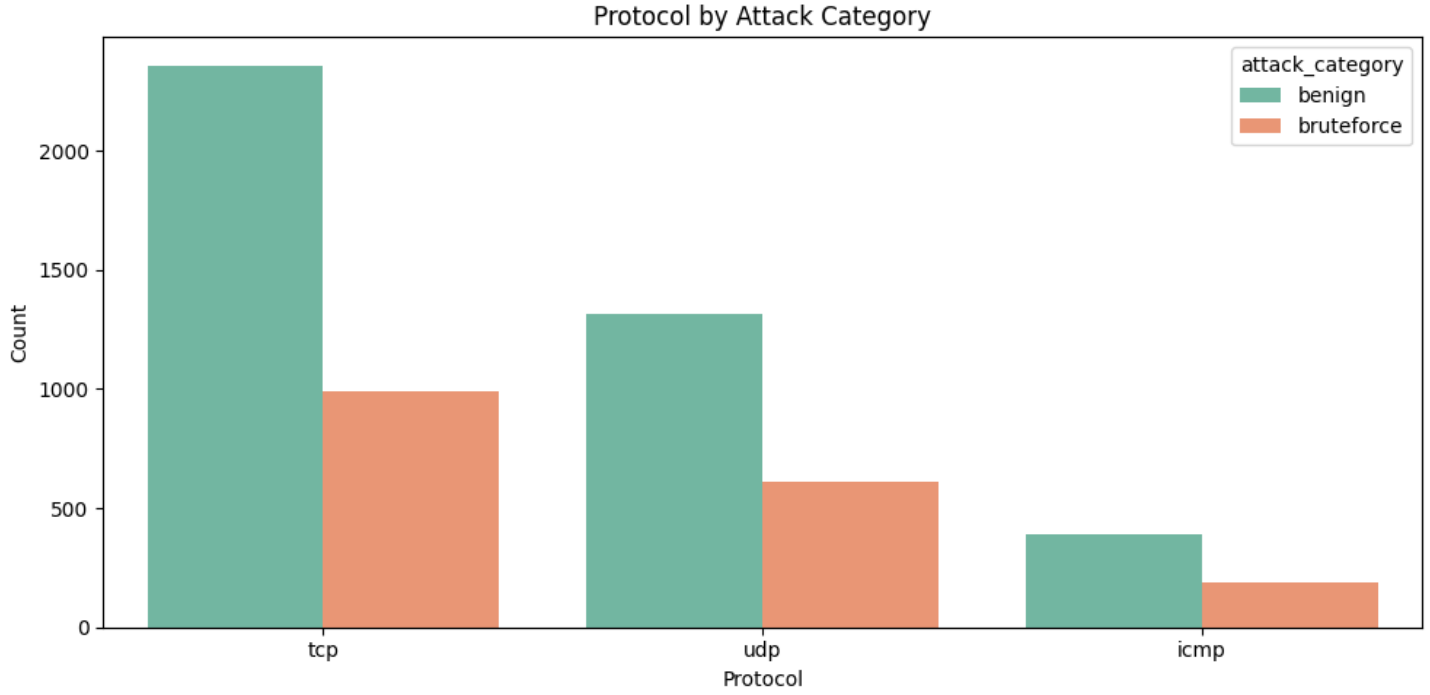


Figure 5 above: Protocol frequency bar chart after lowercasing the columns

The correlation heatmap showed no definite correlations between any pair of features. The "strongest" correlations with average packet size were packet rate (r = 0.3) and connection count (r = 0.2), which are both weak as they are under r = 0.4 to even make a moderate correlation.

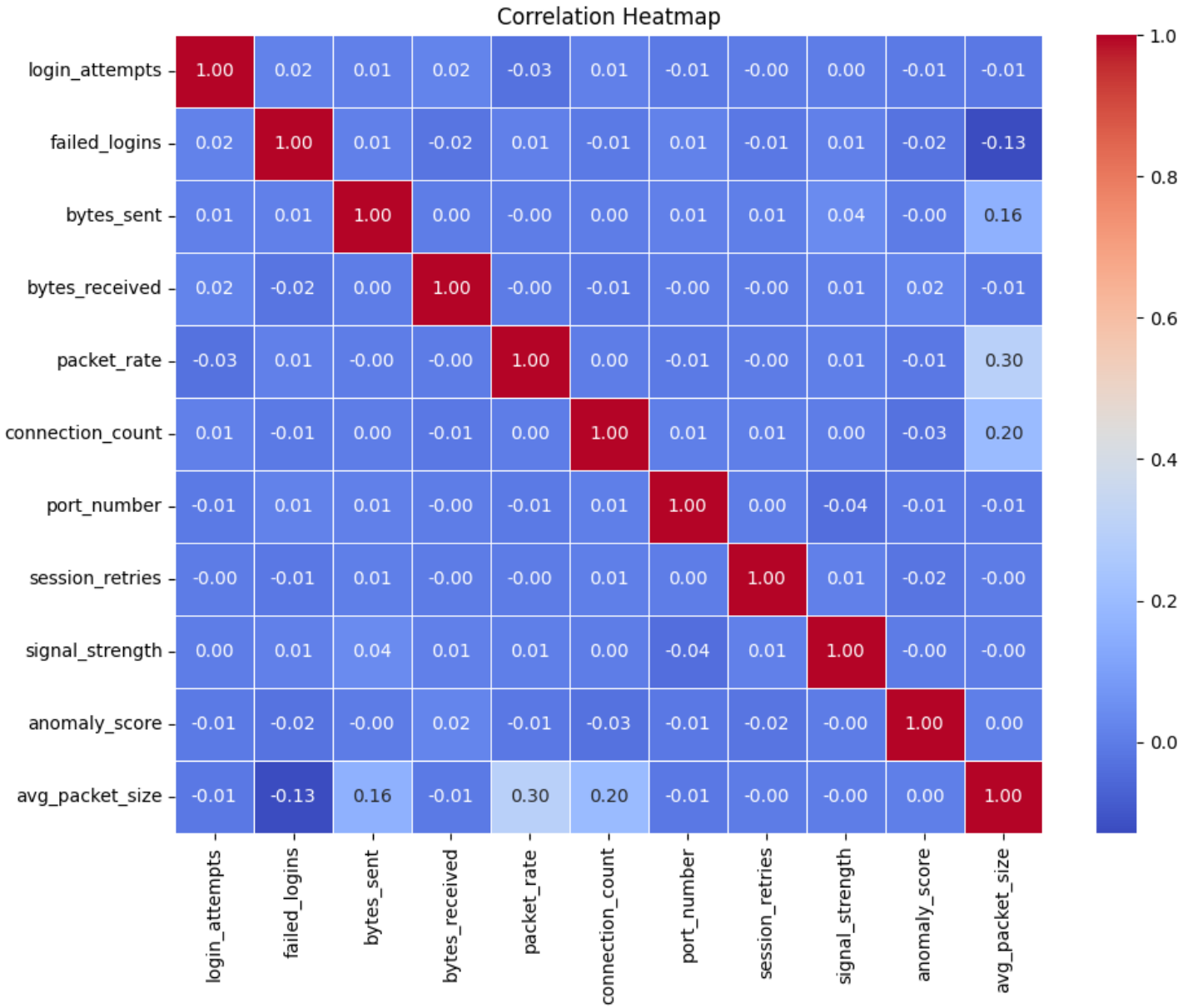


Figure 6 above: Correlation Heatmap

The scatter plot of bytes sent vs bytes received coloured by attack category showed significant overlap between classes with no clear linear boundary separating them. This directly supports the choice of KNN over Logistic Regression as the second classifier as KNN makes no linearity assumption while Logistic Regression would struggle to find a separating line that does not exist.

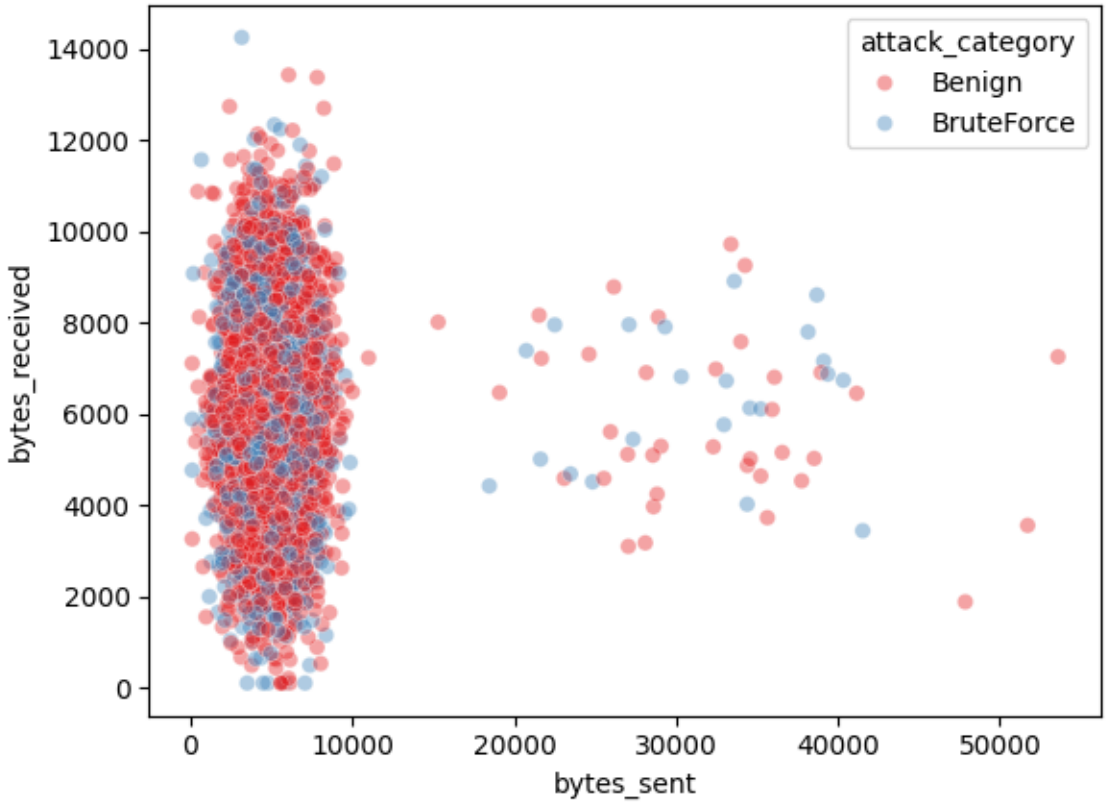


Figure 7 above: Scatter plot bytes_sent vs bytes_received by attack category

Methodology

This report follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework's first five phases: Business Understanding, Data Understanding, Data Preparation, Modelling, and Evaluation.

In Data Preparation, the first step was to fix inconsistencies before missing value percentages are calculated as some inconsistencies would have to be assigned null values. Firstly, all text columns lowercased and stripped of whitespace to merge "TCP" with "tcp" and "UDP" with "udp." 1,109 rows where the recorded failed logins exceeded the total login attempts and 144 rows where the recorded session retries exceeded the total connection count were set to null. Secondly, missing values were handled differently depending on the column's percentage of missing values, the attribute type, and if numerical, whether the data distribution is symmetric or skewed according to their histograms and by comparing the mean and median from the describe() output. As missing values only made up of 2.4% of rows in session retries, those rows were dropped as they have a negligible impact on the model and it is unnecessary to invent values for data, leaving 5856 rows remaining. The rest of the columns had their rows with missing values imputed as they consisted of more than 5% but less than 40% of all of the rows in each of the column. The rows with missing values in protocol were imputed with "tcp" as it was the most frequently occurring value in a categorical attribute. Bytes received was determined to be symmetrical and its missing values were imputed with the mean/average value of 6030.81 to give a complete representation of the known data. Likewise, signal strength's missing values were also imputed with the mean value of -60. After the invalid values were set to null, failed logins still remained skewed and its missing values were imputed with the median integer value of 1 as median values do not get affected by those extreme values.

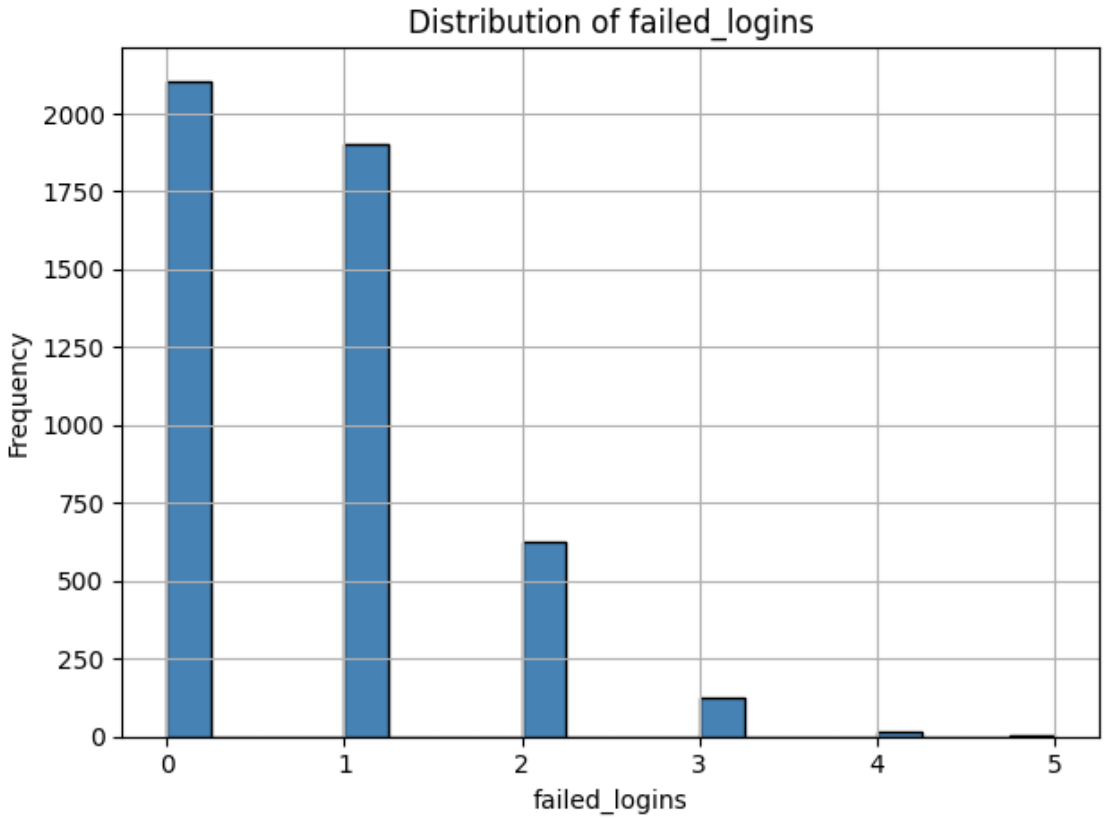


Figure 8 above: Histogram of failed login attempts after imputation

Thirdly, to reduce the effect of skewed data, outliers were capped at its Q1 - 1.5*(Q3 - Q1) and Q3 + 1.5*(Q3 - Q1) using the IQR method, rather than being deleted. This decision was made because network traffic extreme values are genuine session characteristics, not data entry error. However an exception was made for the average packet size column as it is the prediction target and requires its full natural range.

Fourthly, checks were made for zero-variance and near-constants of over 95% in the variance of numeric columns. There were no near-constant columns found but a column for packet session id was determined to be completely unique and was dropped as it is of no predictive value.

Fifthly, three new attributes/columns were added: login failure rate to measure the scale of the brute force attacks, send/receive bytes ratio to measure possible data exfiltration, and session retry which measures rate port scanning. These increase the number of columns to 21.

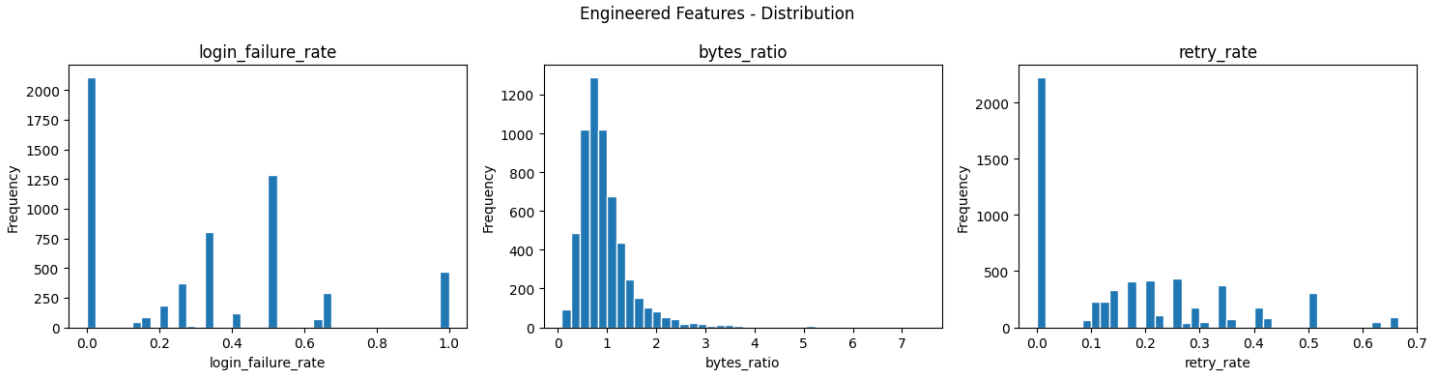


Figure 9 above: Histograms of engineered features

Sixthly, LabelEncoder was applied to all six categorical columns to convert their text data into integers. The classification target was encoded separately so that its numeric codes can be converted back to "Benign" and "BruteForce" for display in the results.

Finally, train and test rows were split on a 80/20 ratio, except for the classification target which kept its 69%/31% class ratio. MinMaxScaler was fitted on training data only to prevent test data leakage to the model.

The Decision Tree algorithm was selected as one of the Classification Models as it can handle this datasets mixed numeric and categorical features and it produces easy to understand interpretable if-then rules and feature importance scores. k-nearest neighbors (KNN) algorithm was selected as the other Classification Models because it provides a simple baseline for a medium, non-linear dataset like this one. Linear Regression was selected for the one of the Regression Models because it assumes a linear relationship between features and average packet size. Decision Tree Regressor was selected for the other Regression Model because it can capture non-linear relationships that linear regression cannot. The Naive Bayes algorithm was ruled out as it is better suited for datasets with independent and mostly categorical features, and this dataset is mostly numerical and has features that correlate with each other. Logistic Regression was also ruled out as unsuitable for assuming a linear boundary between classes which is not this dataset's case as seen by the scatter plot.

As discussed earlier, Recall is the primary evaluation metric in evaluating the classification models as a missed attack (false negative) is far more harmful than a false alarm (false positive) for a financial organisation's

cybersecurity. Prediction models are evaluated using the average of the squared and absolute differences between predicted and actual values, aiming for a low score. The Coefficient of Determination (R2) which measures how much of the variation is explained by the model on a scale from 0 to 1 will also be an evaluation tool, aiming for a high score. When tuning the models, K=11 gave the best accuracy and was selected for KNN, and max_depth=5 gave the best R2 for Decision Tree Regressor.

Results

Classification Results

Metric	Decision Tree (depth=5)	KNN (K=11)
Train Accuracy	0.7047	0.6992
Test Accuracy	0.6860	0.6672
Precision	0.5808	0.5832
Recall	0.6860	0.6672
F1-Score	0.5782	0.5923
CV F1 (5-fold)	0.5752 +/- 0.0040	0.5905 +/- 0.0080

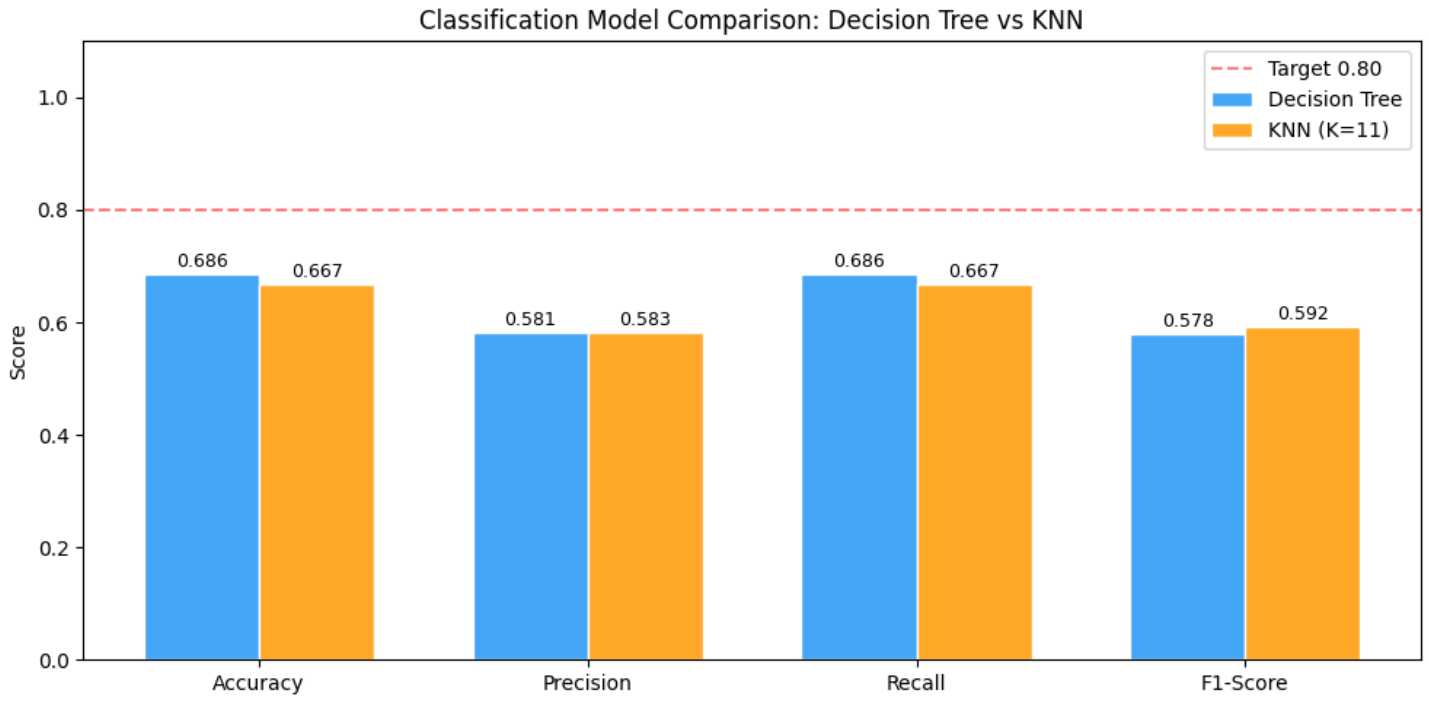


Figure 10 above: Classification Model Comparison bar chart

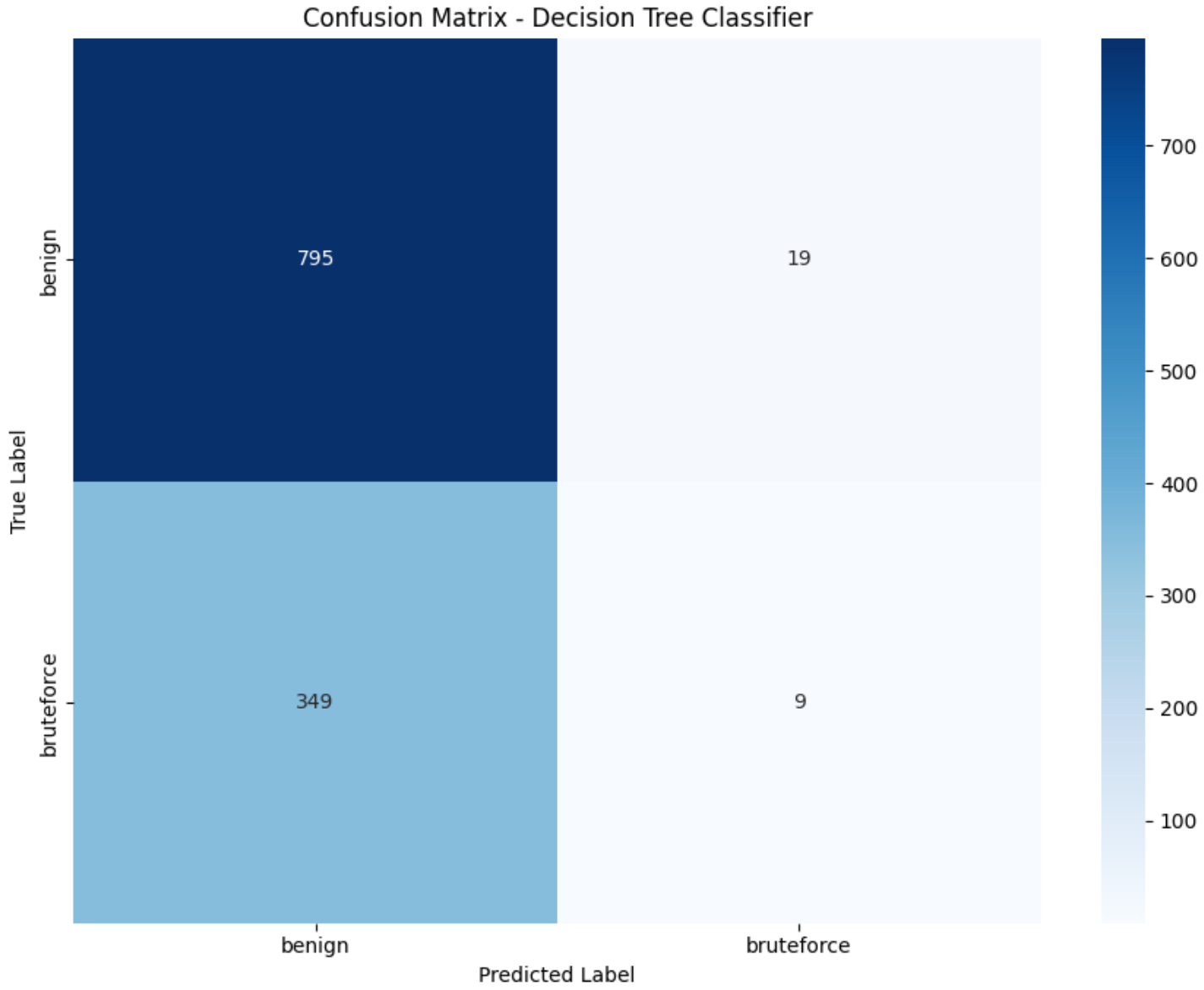


Figure 11 above: Decision Tree confusion matrix

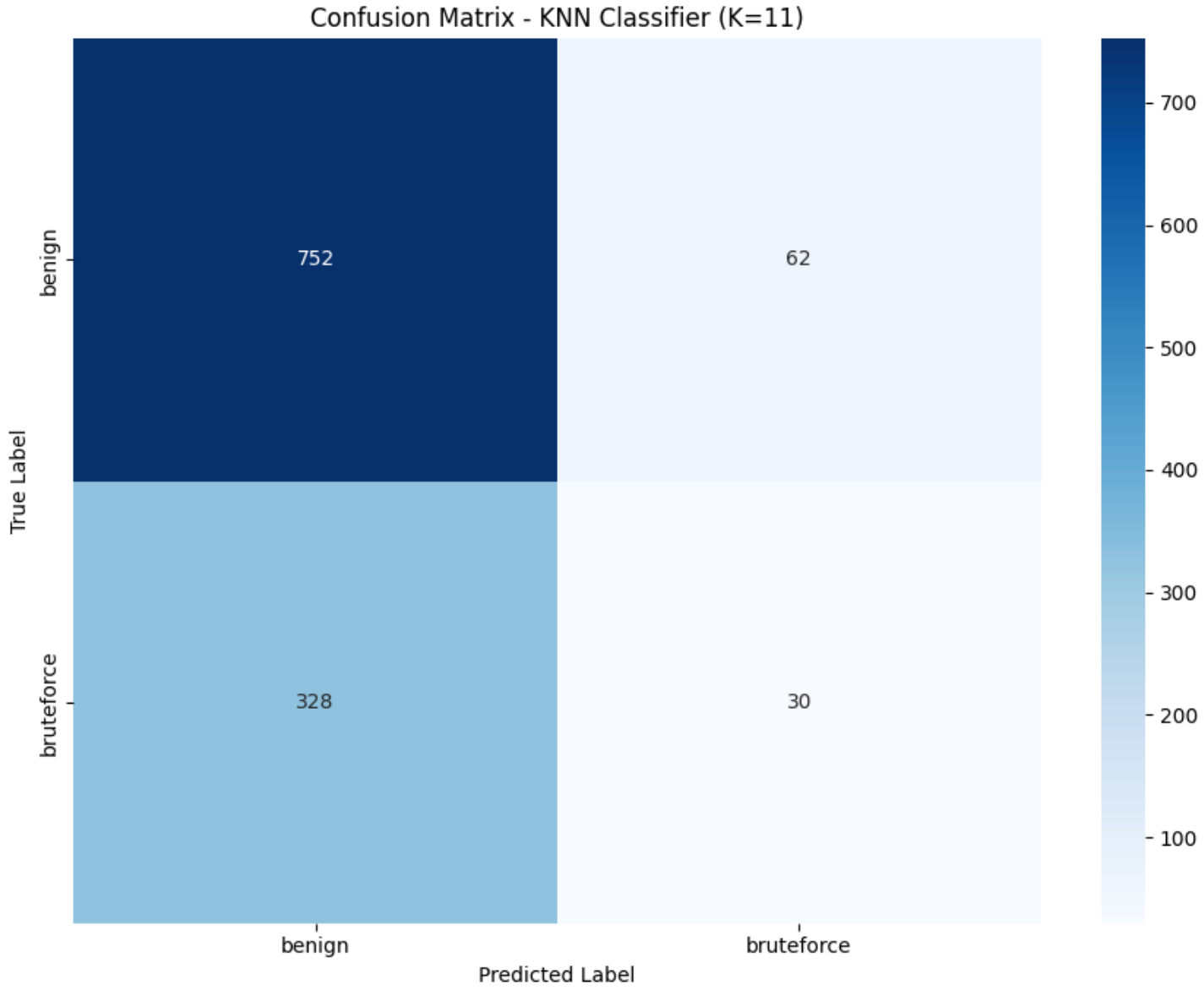


Figure 12 above: KNN confusion matrix

The Decision Tree wins on Accuracy (0.686 vs 0.667) and Recall (0.686 vs 0.667). KNN marginally wins on Precision (0.5832 vs 0.5808) and F1-Score (0.5923 vs 0.5782). Given that a false negative (missed attack) is more harmful than a false positive (false alarm), the Decision Tree is better suited to the business context for having higher recall as it catches more real BruteForce sessions. Cross-validation results are consistent with test set results for both models, confirming neither is significantly overfitting.

Prediction Results

Model	Root mean square deviation	Mean absolute error	Coefficient of Determination
Linear Regression	15.7830	12.2665	0.5480
Decision Tree Regressor	17.1917	13.5788	0.4637



Figure 13 above: Actual vs Predicted scatter plots (both models)



Figure 14 above: Regression Model Comparison bar charts

Linear Regression outperformed the Decision Tree Regressor on all three metrics. Linear Regression can explain 54.8% of the data variation while Decision Tree Regressor can only explain 46.37%. This is somewhat expected as shown in the scatter plots, average packet size is closer to linear than non-linear which is why it performs better with Linear Regression which has a simpler structure that avoids overfitting to noise in the training data that the deeper Decision Tree picks up. The most likely explanation is that the relationship between network features and average packet size is genuinely weak, the correlation heatmap showed the strongest single feature correlation with average packet size was only packet rate at r = 0.30. When the signal is this weak, Linear Regression's simpler structure avoids overfitting to noise that the Decision Tree picks up at deeper splits.

Conclusion

The Decision Tree is the recommended classifier. Despite KNN having a marginally higher F1-score, the Decision Tree catches more real attacks (higher recall) and its feature importance scores give the SOC team actionable intelligence on what to monitor. For a financial organisation, missing an attack is the worst possible outcome.

packet_rate and the engineered bytes_ratio are the two strongest predictors of BruteForce activity. The fact that an engineered feature ranked second confirms that incorporating domain knowledge into the feature set improved model performance meaningfully.

avg_packet_size is not strongly predictable from this feature set. No features had a correlation above r = 0.30 with the target, and both prediction models fell below R2 > 0.70. This appears to be a limitation of the dataset rather than a modelling issue.

References

IBM, n.d. *What is a brute force attack?.* [Online]
Available at: https://www.ibm.com/think/topics/brute-force-attack
[Accessed 27 April 2026].

MITRE ATT&CK, n.d. *Brute Force.* [Online]
Available at: https://attack.mitre.org/techniques/T1110/
[Accessed 27 April 2026].

Link to Publicly accessible Python Notebook with all code, outputs, and markdown explanations

https://colab.research.google.com/drive/1bENmfJZ5B1Kda-ZkjQZVkW_LqYcogExn?usp=sharing